Tallinn University of Technology

Inverse Kinematics (IK)

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Individual variant: RPR8

Course: ATR0030 Robotics

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1 Task

The purpose of this work was to find joint configurations of my assigned manipulator, variant **RPR8**, that place its end effector (EE) at a specified target position. The inverse kinematics search uses the brute-force (generate and test) method: calculate forward kinematics for sampled joint values and keep the configurations that reach the target within a chosen tolerance.

The accepted configurations are plotted together with their coordinate frames. The plot uses `view(3)`, as in my Forward Kinematics report. This changes the camera view only; the search tests EE position and does not impose an EE orientation constraint.

2 Initial Data

My assigned manipulator has the joint sequence Revolute–Prismatic–Revolute. The dimensions are unchanged from the FK and WS reports.

Table 1: Initial link dimensions for variant RPR8

Parameter	Value	Description
L_0	3	Base offset
L_1	5	J1–J2 link length
$L_{2,\max}$	6	Maximum prismatic displacement
L_3	5	J3–EE link length

The joint variables are the angle A at J1, the prismatic displacement L_2 between J2 and J3, and the angle B at J3. At home,

$$A = 0, \qquad B = 0, \qquad L_2 = 6.$$

The FK model gives $P_E = [6\ 5\ 8]$, so the target is

$$\boxed{\text{TARGET} = [6\ 5\ 8].}$$

Home is defined by these joint values, not by a line colour in a plot.

3 Model Parameters

The fixed translation and reorientation vectors are

$$T_{01} = [0\ 0\ L_0], \quad R_{01} = [0\ 1\ 0\ -\pi/2], \quad T_{12} = [L_1\ 0\ 0], \quad T_{3E} = [0\ L_3\ 0].$$

The variable transformations are a local z rotation $r_1(A)$, a translation $t_2 = [0\ 0\ -L_2]$, and a local z rotation $r_3(B)$. After conversion to homogeneous matrices, the chain is

$$^0T_1 = T_{01}R_{01}r_1, \qquad ^0T_2 = {}^0T_1T_{12},$$
$$^0T_3 = {}^0T_2t_2, \qquad ^0T_E = {}^0T_3r_3T_{3E}.$$

The plotted J3 frame includes r_3, so its displayed orientation includes the final revolute-joint angle.

4 RRP1 Adaptations and Search Method

The main method is retained from the professor's RRP1 example described in the earlier exercise discussion. The adaptations are:

- Change the dimensions to $L_0 = 3$, $L_1 = 5$, $L_{2,\max} = 6$ and $L_3 = 5$.
- Make J2 prismatic and J3 revolute: replace the final variable translation with t_2, and apply angle B at J3.
- Use the initial $-\pi/2$ rotation about y from the RPR8 FK model and select the known FK target [6 5 8].

The program uses $N = 20$ samples per variable, giving $20^3 = 8000$ tested combinations. Both angles are sampled at $0°, 18°, \ldots, 342°$, and L_2 is sampled uniformly from 6 down to 0. The expression `(0:N-1)*2*pi/N` avoids duplicating $0°$ and $360°$. For $N = 20$, it preserves the angular grid used in the earlier IK code, without linking angular coverage to the position tolerance.

A configuration is accepted when

$$|x_E - 6| < 0.05, \qquad |y_E - 5| < 0.05, \qquad |z_E - 8| < 0.05.$$

This is a tolerance on each coordinate, not a Euclidean distance limit. The matrix `configs` stores one accepted configuration per column, with rows A in degrees, B in degrees, and L_2 in the model's length units.

Unlike the WS exercise, this search does not apply the workspace collision restrictions. It identifies position matches, not certified collision-free motions. A finite grid can miss solutions between samples, and its accepted count depends on resolution and tolerance.

5 MATLAB Program

The complete script below contains minimal comments describing the RPR8 changes. It saves the actual configuration matrix and the figure when run in MATLAB with the required robotics transformation and plotting functions available.

```
clear; clc; close all

%% Jordan Brodie - RPR8 Inverse Kinematics
% RRP1 -> RPR8: L2 is prismatic; L3 is fixed.
L0=3; L1=5; L2max=6; L3=5;
T01H=trvec2tform([0 0 L0]);
% Use the FK reorientation at J1 instead of RRP1's R12.
R01H=axang2tform([0 1 0 -pi/2]);
T12H=trvec2tform([L1 0 0]);
T3EH=trvec2tform([0 L3 0]);

%% Search settings
N=20;
d=0.05;
rangeA=(0:N-1)*2*pi/N;
rangeB=(0:N-1)*2*pi/N;
rangeL=linspace(L2max,0,N);
% Target from FK: A=0, B=0, L2=6 gives EE=[6 5 8].
TARGET=[L2max L3 L0+L1];
configs=zeros(3,0);
cnt=0;
figure('Color','w');
hold on; grid on; axis equal
xlabel('x'); ylabel('y'); zlabel('z');
colours=lines(N^2);

%% Generate and test
for L2=rangeL
    for B=rangeB
        for A=rangeA
            r1h=axang2tform([0 0 1 A]);
            % Move translation to J2 and rotation B to J3.
            t2h=trvec2tform([0 0 -L2]);
            r3h=axang2tform([0 0 1 B]);
            J1=T01H*R01H*r1h;
            J2=J1*T12H;
            J3=J2*t2h;
            EE=J3*r3h*T3EH;
            P1=tform2trvec(J1);
            P2=tform2trvec(J2);
            P3=tform2trvec(J3);
            PE=tform2trvec(EE);
            if all(abs(PE-TARGET)<d)
                cnt=cnt+1;
                configs(:,cnt)=[rad2deg(A); rad2deg(B); L2];
                P=[0 0 0; P1; P2; P3; PE];
                colour=colours(mod(cnt-1,size(colours,1))+1,:);
                plot3(P(:,1),P(:,2),P(:,3),'-o', ...
                    'Color',colour,'LineWidth',1.5);
                labels={'J0','J1','J2','J3','EE'};
                for k=1:5
```

```
                    text(P(k,1)+0.15,P(k,2),P(k,3),labels{k});
                end
                frames={J1,J2,J3*r3h,EE};
                for k=1:4
                    H=frames{k};
                    plotTransforms(tform2trvec(H),tform2quat(H));
                end
            end
        end
    end
end
plot3(TARGET(1),TARGET(2),TARGET(3),'kp', ...
    'MarkerSize',12,'LineWidth',1.5);
title(sprintf('Jordan Brodie RPR8: %d sampled configurations',cnt));
view(3) % Match the FK camera view.
fprintf('Sampled configurations found: %d\n',cnt);
fprintf('Rows of configs: A (deg), B (deg), L2\n');
format shortG
disp(configs)

%% Save the actual MATLAB results beside this script
outDir=fileparts(mfilename('fullpath'));
if isempty(outDir), outDir=pwd; end
save(fullfile(outDir,'IK_results.mat'),'configs','cnt','TARGET','N','d');
writematrix(configs,fullfile(outDir,'configs.csv'));
exportgraphics(gcf,fullfile(outDir,'IK_figure.png'),'Resolution',300);
```

6 Simulation Results

The MATLAB run found two sampled configurations for the target [6 5 8]. Figure 1 shows the actual Command Window output. Each column of `configs` is one accepted configuration; the rows are A in degrees, B in degrees, and L_2.

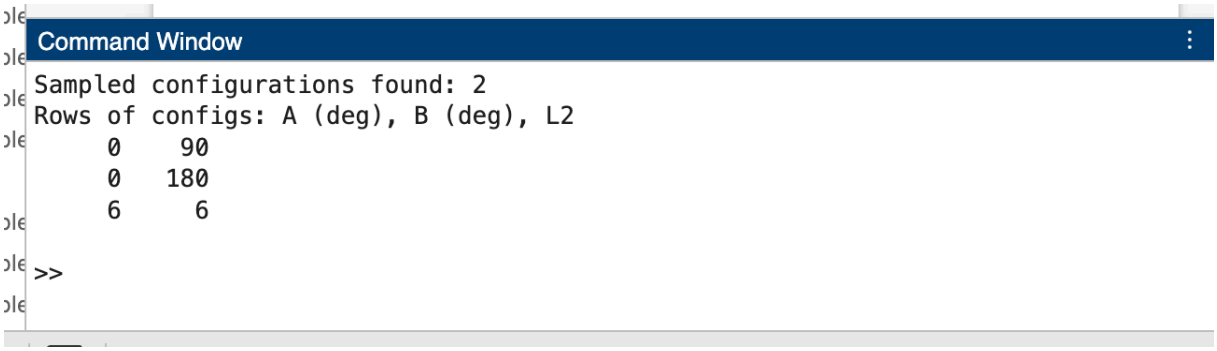


Figure 1: MATLAB output showing the two accepted sampled configurations.

The configuration matrix is

$$\text{configs} = \begin{bmatrix} 0 & 90 \\ 0 & 180 \\ 6 & 6 \end{bmatrix}.$$

The first column is the home configuration: $A = 0°$, $B = 0°$ and $L_2 = 6$. The second column has $A = 90°$, $B = 180°$ and the same prismatic displacement $L_2 = 6$.

Both configurations reach the target within the specified coordinate tolerance. The intermediate joint positions and link orientations differ, while the end-effector position is the same. The search tests position only, so it does not require the two end-effector orientations to match.

The reported count is the number of configurations accepted on this particular grid. It does not by itself prove that there are only two mathematical solutions.

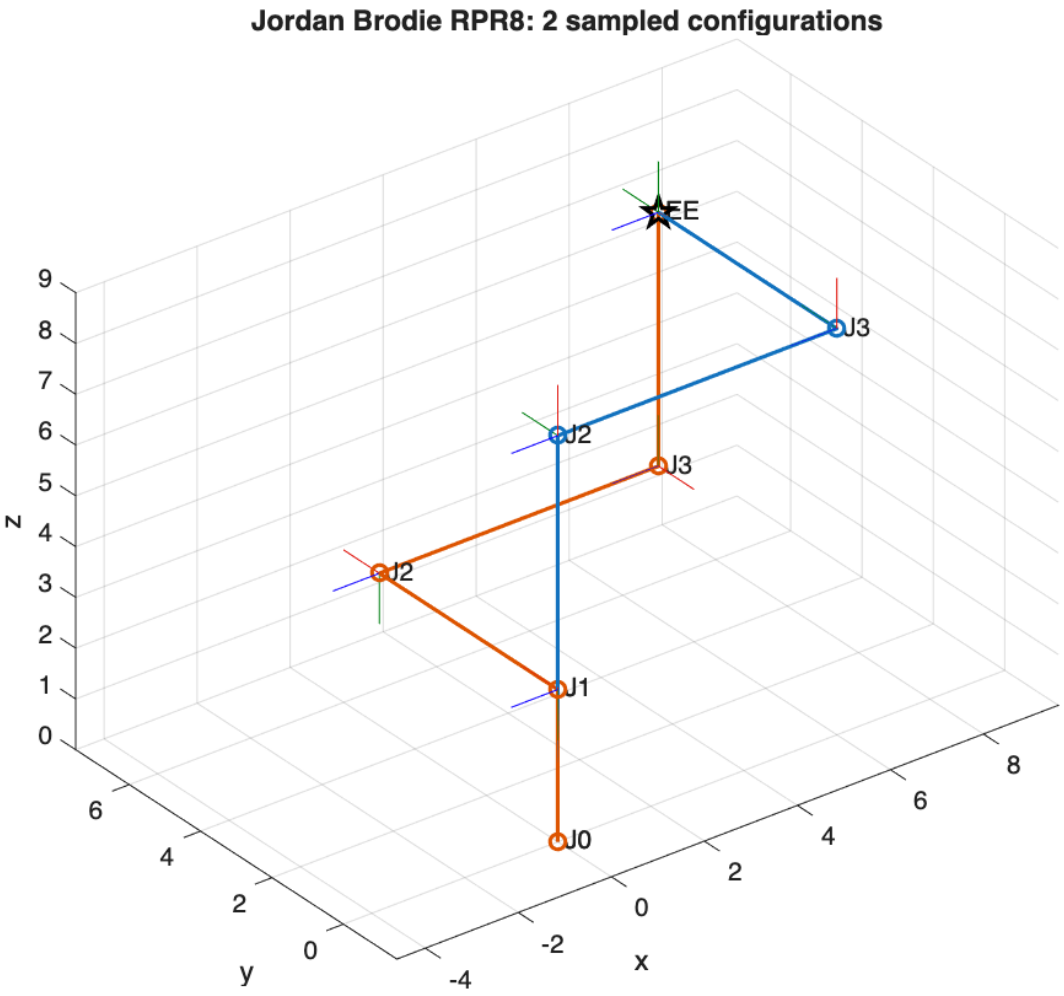


Figure 2: MATLAB plot of the two sampled RPR8 configurations for target [6 5 8], shown from the selected camera angle.

In Figure 2, the blue links show the home configuration and the orange links show the second sampled configuration. The black target marker coincides with the EE position. The short red, green and blue lines represent local coordinate axes.

7 Conclusions

I am surprised that only two sampled configurations were found for my target point. I would like to study other target locations and see how many different configurations the program finds. I think it is very interesting to see how the positions are calculated in the digital world.

I also understand that the two configurations found on this sampling grid do not establish the total number of mathematical solutions. Changing the target, the sampling resolution or the tolerance could change the number of accepted configurations.